Related Rates

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

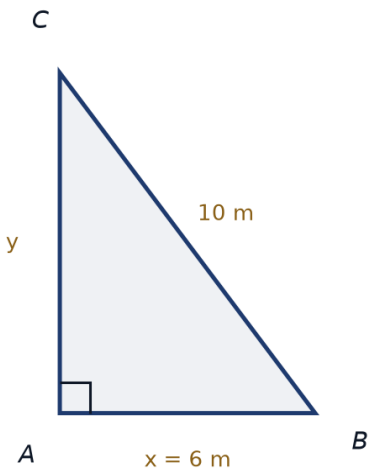
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Setting up related-rates problems**

A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?

2 **[Foundational]** • **Target: Setting up related-rates problems**

A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?



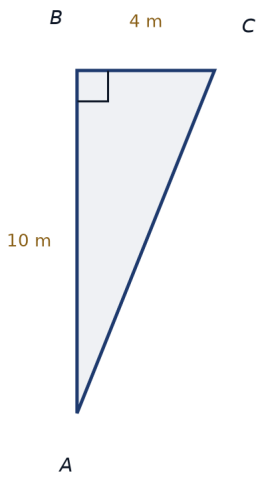
3 **[Foundational]** • **Target: Volume and other rates**

Air is pumped into a spherical balloon at 100 cm^3/s. Find the rate of change of the radius when $r = 5$ cm, to three decimal places.

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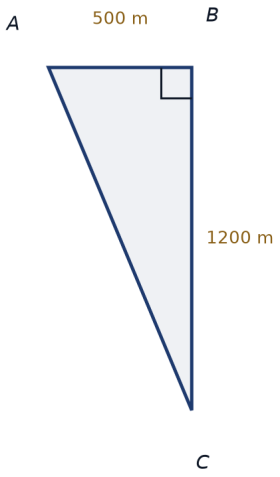
4 [Foundational] • Target: Volume and other rates

Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of 2 m^3/min. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5$ m.



5 [Foundational] • Target: Angle and distance rates

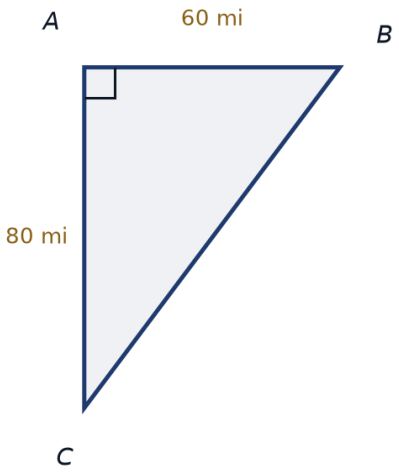
A camera tracks a rocket launching vertically, 500 m from the launch pad. When the rocket is 1200 m high and rising at 200 m/s, how fast is the camera's angle of elevation increasing? Give the answer in radians/s, to four decimal places.



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6 [Foundational] • Target: Angle and distance rates

Two cars leave an intersection at the same time, one heading north at 40 mph and the other heading east at 30 mph. How fast is the distance between them increasing after 2 hours?



7 [Foundational] • Target: More related rates

A streetlight is 6 m high. A 1.8 m tall person walks away from it at 1.5 m/s. Using similar triangles, find how fast the length of the person's shadow is growing.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

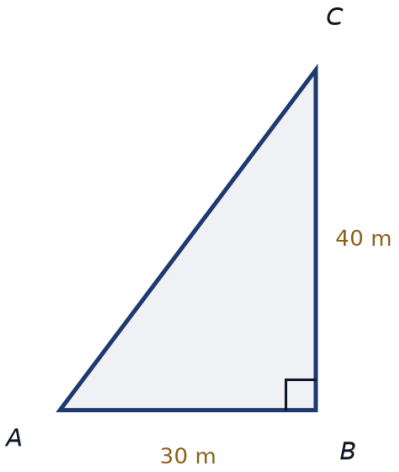
8 [Intermediate] • Target: More related rates

The volume of a cube is increasing at $12\ \text{cm}^3/\text{s}$. Find the rate of change of the side length when the side is $3\ \text{cm}$.

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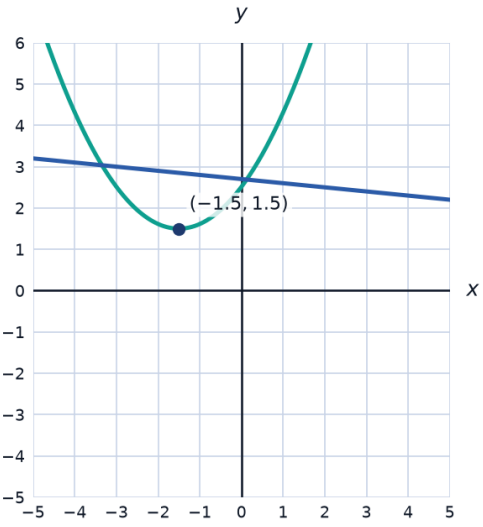
9 [Intermediate] • Target: Rates involving trigonometric relationships

A kite is flying at a constant height of 40 m, moving horizontally away from the person holding the string at 3 m/s. Find the rate at which the string length is increasing when the kite is 30 m horizontally from the person.



10 [Intermediate] • Target: Rates involving trigonometric relationships

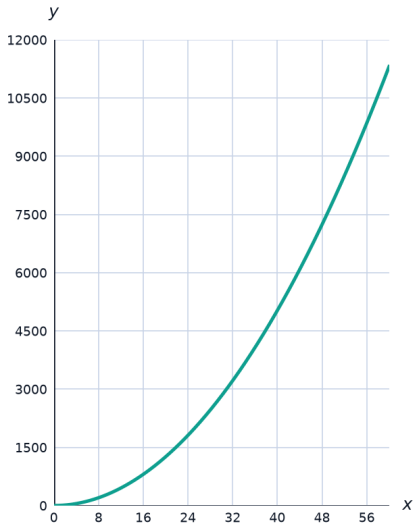
A conical pile of sand has its height always equal to its base radius. Sand is added at $10\ \text{m}^3/\text{min}$. Find the rate of change of the height when $h = 5$ m, using $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.



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11 [Intermediate] • Target: Visualizing a related-rates scenario

The graph shows $A(r) = \pi r^2$, the area of the oil spill from the first problem, as a function of radius alone (not time). Explain why $\frac{dA}{dt}$ still depends on $\frac{dr}{dt}$ even though this graph has no time axis.



12 [Intermediate] • Target: Related rates with area and perimeter together

A rectangle's length is always twice its width. If the width is increasing at 3 cm/s, find the rate of change of the area when the width is 5 cm.

13 [Intermediate] • Target: Related rates with area and perimeter together

For the same rectangle (length $= 2 \times$ width), find the rate of change of the perimeter when the width is increasing at 3 cm/s.

14 [Intermediate] • Target: A two-stage related-rates problem

A spherical balloon is being inflated so that its surface area $S = 4\pi r^2$ increases at 12π cm^2/s. Find $\frac{dr}{dt}$ when $r = 3$ cm, then use it to find $\frac{dV}{dt}$ at that instant.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

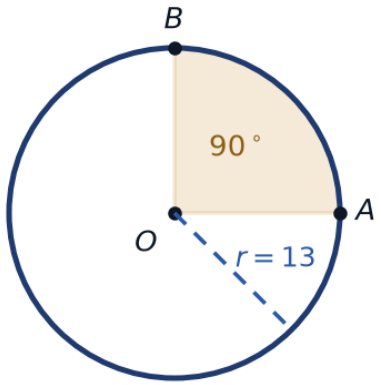
15 [Challenging] • Target: Related rates with area and perimeter together • Cross-Lesson: 15.01 Double Integrals

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

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16 [Challenging] • Target: A two-stage related-rates problem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.



17 [Challenging] • Target: Setting up related-rates problems • Cross-Lesson: 2.01 Evaluating Limits

Level 1 · Foundations Find the particular solution to $\frac{dy}{dx} = \frac{3x^2}{y}$ subject to $y(0) = 4$.

18 [Challenging] • Target: Volume and other rates • Cross-Lesson: 2.02 Continuity

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

19 [Challenging] • Target: Angle and distance rates • Cross-Lesson: 3.01 The Derivative from First Principles

Level 1 · Foundations Find $\frac{dy}{dx}$ in terms of t for the parametric curve $x = 2\cos t$, $y = 5\sin t$ for $t \in (0, \pi)$.

20 [Challenging] • Target: More related rates • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Setting up related-rates problems

A cone of fixed volume is to be made with minimal lateral surface area. Optimal half-angle α satisfies:

a. $\sin\alpha = 1/\sqrt{2}$

b. $\tan\alpha = \sqrt{2}$

c. $\cos\alpha = 1/2$

d. $\alpha = \pi/3$

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22 [Foundational] • Target: Volume and other rates

Related rates: sphere volume $V = (4/3)\pi r^3$. If $dr/dt = 2$ cm/s at $r = 3$ cm, $dV/dt =$

a. 72π cm³/s

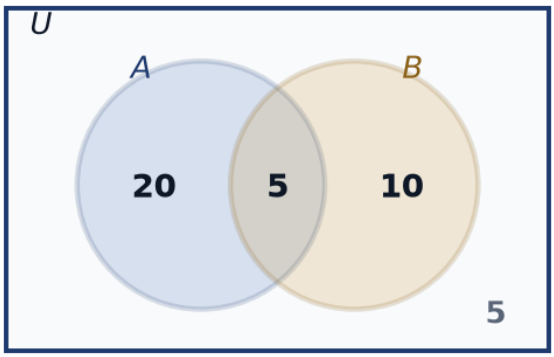
b. 36π cm³/s

c. 12π cm³/s

d. 24π cm³/s

23 [Foundational] • Target: Angle and distance rates

Volume of the solid of intersection of two cylinders $x^2 + z^2 \le 1$ and $y^2 + z^2 \le 1$:



a. $\frac{8}{3}$

b. $\frac{16}{3}$

c. π

d. 4

24 [Foundational] • Target: More related rates

Related rates: a 10-m ladder slides down a wall. When the base is 6 m from the wall and moving away at 2 m/s, the top is moving at:

a. $-\frac{3}{2}$ m/s

b. $-\frac{3}{4}$ m/s

c. $\frac{3}{2}$ m/s

d. $\frac{4}{3}$ m/s

25 [Foundational] • Target: Rates involving trigonometric relationships

Related rates: water drains from a conical tank (height $H = 10$m, radius $R = 5$m) at 2 m³/min. When depth $h = 4$ m, $dh/dt =$

a. $-\frac{2}{4\pi}$ m/min

b. $-\frac{1}{2\pi}$ m/min

c. $-\frac{2}{\pi}$ m/min

d. $-\frac{8}{\pi}$ m/min

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26 [Foundational] • Target: Visualizing a related-rates scenario

Optimize area of rectangle with perimeter 20: max area $=$

a. 25

b. 20

c. 50

d. 100

27 [Foundational] • Target: Related rates with area and perimeter together

Related rates: if $A = r^2\pi$, then $\frac{dA}{dt} =$

a. $2\pi r$

b. $2\pi r\frac{dr}{dt}$

c. $\pi r^2\frac{dr}{dt}$

d. $2\pi\frac{dr}{dt}$

28 [Foundational] • Target: A two-stage related-rates problem

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$

d. $\pi(\sqrt{2} - 1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Setting up related-rates problems

An open-top box with a square base must have volume $8\ \text{m}^3$. To minimise the surface area $S(x) = x^2 + \frac{4(8)}{x}$, the optimal base side x satisfies:

a. $x = \sqrt[3]{16}$

b. $x = \sqrt[3]{8}$

c. $x = \frac{8}{2}$

d. $x = \sqrt{16}$

30 [Intermediate] • Target: Volume and other rates

Related rates: ladder 10 m against wall. Base slides at 1 m/s when base is 6 m. Rate of top:

a. $-3/4$ m/s

b. $3/4$ m/s

c. -1 m/s

d. $-4/3$ m/s

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31 [Intermediate] • Target: Angle and distance rates

Fence three sides of rectangle next to wall (wall is 4th side). Area $= xy = A$, perimeter $= x + 2y = L$. Max area:

a. $L^2/8$

b. $L^2/4$

c. $L^2/6$

d. $L^2/2$

32 [Intermediate] • Target: More related rates

Two vectors are anti-parallel if angle between them is:

a. 0

b. $\pi/4$

c. π

d. $\pi/2$

33 [Intermediate] • Target: Rates involving trigonometric relationships

Optimal rectangle of fixed perimeter P has max area when:

a. length $= 2 \times$ width

b. it is a square

c. width $= P$

d. length $= P/4$

34 [Intermediate] • Target: Visualizing a related-rates scenario

Surface area revolving $x = t^2, y = 2t$ $(0 \le t \le 3)$ about x-axis:

a. $\pi(2 + 2\sqrt{2})$

b. $24\pi\sqrt{2}$

c. $8\pi(10\sqrt{10} - 1)/3 \cdot \ldots$

d. $48\pi\sqrt{2}$

35 [Intermediate] • Target: Related rates with area and perimeter together

Find the area inside $r = 2\cos(\theta)$ and outside $r = 1$.

a. π

b. $\sqrt{3} + \frac{2\pi}{3}$

c. $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$

d. $\frac{\pi}{3}$

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36 [Intermediate] • Target: A two-stage related-rates problem

Area below x-axis from -2 to 0 of $f(x) = x^2 - 4$:

a. $-\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{16}{3}$

d. 4

37 [Intermediate] • Target: Setting up related-rates problems

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi\int(R-r)^2dx$

b. $\pi\int(R^2-r^2)dx$

c. $2\pi\int(R-r)dx$

d. $\pi\int R^2dx-\pi\int r^2dx$ (same)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume and other rates • Cross-Lesson: 2.02 Continuity

Centroid $\bar{x}$ of region R with area A:

a. $\frac{a+b}{2}$

b. $\frac{1}{A}\int_a^b f(x)\,dx$

c. $\frac{1}{A}\int_a^b x[f(x)-g(x)]\,dx$

d. $A/2$

39 [Challenging] • Target: Angle and distance rates • Cross-Lesson: 3.01 The Derivative from First Principles

Find the area of the region inside both $r = 8\sin(\theta)$ and $r = 8\cos(\theta)$.

a. 8π

b. $-32+16\pi$

c. 4π

d. $-16+8\pi$

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40 [Challenging] • Target: More related rates • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Shell method: volume when shell radius $= x$, height $= f(x)$:

a. $2\pi\int_a^b xf(x)dx$

b. $\int_a^b f(x)dx$

c. $\pi\int_a^b x^2 f(x)dx$

d. $\pi\int_a^b f(x)^2 dx$

41 [Challenging] • Target: Rates involving trigonometric relationships • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Volume of a paraboloid $z = x^2 + y^2 \le 4$:

a. 16π

b. π

c. 4π

d. 8π

42 [Challenging] • Target: Visualizing a related-rates scenario • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:

a. $f > g$ and $f < g$ alternates

b. $f > 0$ and $g < 0$

c. $f(x) \ge g(x)$ on $[a, b]$

d. always

43 [Challenging] • Target: Related rates with area and perimeter together • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the area enclosed by the limaçon $r = 2\cos(\theta) + 5$.

a. 25π

b. 29π

c. 54π

d. 27π

44 [Challenging] • Target: A two-stage related-rates problem • Cross-Lesson: 6.02 Optimization and Curve Sketching

Optimal Lagrange multiplier problem: maximise $f(x, y)$ subject to $g(x, y) = 0$ gives condition:

a. $ablaf = 0$

b. $ablaf = \lambda ablag$

c. $ablag = 0$

d. $f = g$

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45 [Challenging] • Target: Setting up related-rates problems • Cross-Lesson: 7.01 Integration by Substitution

Surface area of cone with radius r, slant height l:

a. $\pi r^2 l$

b. $\pi r l$

c. $2\pi r l$

d. $\pi r(r + l)$

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